

Management Information Systems

CSC 448

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COURSE OUTLINE

- Definition, types of information systems, requirements
- Systems Development life cycle (SDLC): Historical Perspective, traditional SDLC, Potential Strengths and weaknesses of Traditional SDLC
- Information systems Development: the system approach, strategic information system, Business process re-engineering , planning approaches, process modeling, data modeling, object oriented systems development, prototyping, Software engineering and formal methods, expert systems
- **Delivery strategy:** Use PowerPoint presentations/report, Give made, Students can analyze some systems in the faculty or in some organizations practical assignment in groups, Groups to present and discussion and question in town e.g. Navrongo health research , hospitals etc., Students may develop a prototype of some systems in the university or any organisation

Management Information Systems

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Management information system, or MIS, broadly refers to a computer-based system that provides managers with the tools to organize, evaluate and efficiently manage departments within an organization.



Recommended Books:

- Haag, S., Cummings, M., and McCubbrey, (2004) *Management Information Systems for the Information Age*. New York: McGraw-Hill.
- Oz, E., (2004) *Management Information Systems*. America: Pennsylvania State University.



Essential Books

- K., C., Laudon and J., P., Laudon (2007), '10th edn', *Management Information Systems: Managing the Digital Economy*. New Jersey: Pearson Education.
- Turban et al (2008) '6th edn' *Information Technology for management Transforming Organisations in the Digital Economy*. River Street: Quebecor World.



LECTURE 1:

Lecture Objectives

- At the end of this lecture students should be able :
- To explain why information systems are of such importance to businesses of today.
- To explain how data is processed into useful information for decision making.



Lecture objectives

cont:

- Explain the components of management information systems.
- Describe the stages of processing information
- Explain the role of information systems in functional business areas



Lecture objectives

cont:

- Explain the differences between, data, information and systems.
- Discuss the role of information systems in business sectors.
- Explain the role of information systems in business transformation.



What is data?

- **Data** are streams of raw facts representing events happening in an organisation.
- They are the raw materials for information production.
- Data may be made up of numbers, words, and images.
- These collections of facts are mainly for decision making



Definition of information

- **Information** :on the other hand is data that has been processed into a meaningful and useful form in a context for a decision making.
- It is a manipulated data that leads to greater understanding of a situation.
- Information is knowledge obtained from a study, experience or instructions.



Characteristics of useful information:

- information should pertain to the problem at hand(Relevant)
- Information shouldn't be partially presented(Should be Complete)
- Information shouldn't be erroneous (Accuracy)



Characteristics of useful information:

- Decision should base on the latest information.
Fact changes from time to time (Timely)
- Information ought to be obtained economically,
especially in business(Cost Effective)



Understanding Systems:

- A system is a collection of components that work together to achieve a common goal.
- A system accepts Inputs, process it and produce an output.



Information system:

- Hence **information system** is all components that work together to process and produce information.
- In an organisation, an information system consists of data, hardware, software, networks, people, and procedures.



Components of Information systems:

- Data: Inputs
- Hardware: Computer(motherboard & processor) and its peripheral equipment: input, output, and storage devices
- Software: Set of instructions for computers as to how to input data, how to process, how to display and store information



Components of Information systems:

- Telecommunication(Networks): Hardware and software that ensures transmission and reception of numbers, text, pictures, sound, and animation in the form of electronic data



Components of Information systems:

- People: Information experts and users, in charge of analysing organisational information requirements, design and construct information systems.
- The people are also involved in writing computer programmes, operate the hardware and maintain the software



Components of Information systems

- Procedures: Rules and regulations with regard to optimal and secure operation in processing data.



The stages of Data processing:

- **Input and transaction recordings:** Entering data into the Information system:
- **Processing of data:** Changing and manipulation stage.
- **Display of information:** visual display of information



The stages of Data processing:

- **Storage** : information transferred to magnetic and optical discs and tapes



Information systems in business transformation:

- **Essential Business Tools:**
- Cell phones
- email
- online conferencing
- international conferencing over the internet
- Dot-com internet sites
- Zoom meeting



Information systems in business transformation:

- **Engine for globalise opportunities:**
- International market on the web 24/7
- Google and eBay- reduction in operation cost
- Management of production worldwide



Information systems in business transformation:

- **Information systems aiding the emergence of digital firms:**
- Firms like Dell Computers and Cisco Systems are near digital with the help of internet.



Management Information systems and Strategic Business objects:

- Why information systems so important to business?
- Information system has become the bedrock of achieving strategic business objectives in recent years.



Management Information systems and Strategic Business objects:

- **1) Ensures operational excellence:**
 - e.g. the retailLink system of Wal-Mart/ Asda
- **2) Encourage development of new products, services and new business models:**
 - E.g. In 2006, Spotify launched a free online music service to rival freely available online music and pirated music.



Management Information systems and Strategic Business objects:

- **3) It helps to build and maintain good Customer and Supplier Relationship:**
- A) Customer relationship E.g. A hotel tracking customers' preferences
- B) Supplier relationship: e.g. A pharmacy tracking the quality of drugs supplied by manufacturing firms.



Management Information systems and Strategic Business objects:

- **4) Improvement in decision making :**
 - E.g. Web-based digital dashboard to provide real time information on customer
- **5) Competitive advantage:** e.g.. In 2009, WhatsApp launches a free messaging App which rivalled the traditional networks' Short Message Service (SMS).



Management Information systems and Strategic Business objectives:

- **6) Survival :** Citibank –ATM(1977), In 1977, the chairman of Citibank took a huge risk, spending more than \$100 million to install ATMs all over New York City. That investment paid off the following January when a huge blizzard hit New York, dumping 17 inches of snow on the city. Banks were closed for days; meanwhile, ATM use increased by 20 percent.
- Covid 19 is a recent example. A boost for online business/teaching/banking.



Information systems in functional business areas:

Human Resources :

- **INPUT:**
- Human resource planning
- **PROCESSING:**
- Recruiting and workforce management
- Compensation and benefits.
- **OUTPUTS:**
- Government reporting and labour relations support



Information systems in functional business areas:

- **Accounting:**
- **Finance:**
- **Marketing:**
- Try and search and read on this systems- inputs, processing and output.



Information Systems in Business Sectors:

- Manufacturing: optimises productivity:
- E.g. inventory control system:
- Service: online reservation for airlines and ATM.
- Retailing: big stores operating as a single small stores worldwide.



Information Systems in Business Sectors:

- Development of new business: Credit history organisation, Comparison sites, Amazon.com, eBay
- Government: Tax authorities having tax forms via the web



CASE QUESTIONS:

- What external factors affect international operations at UPS – United Parcel Service? How do these factors cause UPS to adjust its operations?
- Explain how International Shipments Processing System (ISPS) facilitates the ability of UPS to ship packages internationally.
- Describe the role that the Delivery Information Acquisition Device (**DIAD**) plays in UPS business processes.



CASE QUESTIONS

- How important is information to the global expansion of UPS? What advantages does UPS gain by carefully capturing information?
- Discuss the role of volume in the business activities of UPS.



UPS CASE STUDY: